Machine Learning – Regression:

In Machine Learning Domain, under supervised learning for regression method, (prediction output based on numerical values), for the dataset,"50_Startups.csv", first created model using **Multiple Linear Regression** Algorithm, where R2_score is **0.9358**, as the accuracy is **93%**.

Further trained the model using **Support Vector Machine** Algorithm, with hyper tuning parameters, to check for the greater R_score value, which is listed below.

Hyper tuning Parameters	Kernel=Linear (R2_score)	Kernel=rbf (Radial Bias Function) (R2_score)	Kernel=Poly (R2_score)	Kernel=Sigmoid (R2_score)
C=10	-0.0396	-0.0568	-0.0536	-0.0547
C=100	0.1064	-0.0507	-0.0198	-0.0304
C=500	0.5928	-0.0243	0.1146	0.0705
C=1000	0.7802	0.0067	0.2661	0.1850
C=2000	0.8767	0.0675	0.4810	0.3970
C=3000	0.8956	0.1232	0.6370	0.5913

From the above table, the model performed well for **C=3000** and kernel =**" Linear",** where the accuracy is **89%.**

Next, created model using **Decision Tree** algorithm with the parameter as follows:

S.No.	**Criterion**	**Splitter**	**R2_score**
1.	squared_error	best	0.9268
2.	squared_error	random	0.7521
3.	absolute_error	best	0.9514
4.	absolute_error	random	0.8028
5.	poisson	best	0.9164
6.	poisson	random	0.9058

From the above table, the criterion=" **absolute_error"** with splitter=" **best"** performed well as it has the highest R2_score with accuracy of **95%**.

Further created model using **Random Forest** algorithm with the parameter as follows:

S.No.	Criterion	Estimators	R2_score
1.	squared_error	n=10 n=50 n=100	0.9252 0.9446 0.9460
2.	absolute_error	n=10 n=50 n=100	0.9281 0.9401 0.9459
3.	poisson	n=10 n=50 n=100	0.9304 0.9463 0.9413

From the above table, the criterion=**" poisson"** with estimator value **"n=50"** performed well as it has the highest R2_score with accuracy of **94%**.

Overall while comparing the algorithms Multiple Linear Regression, Support Vector Machine, Decision Tree and Random Forest for the dataset "50_Startups.csv", **Decision Tree** algorithm performed well, that is it created the best model with high accuracy score.